Completeness: consistent implies satisfiable

The Open Logic Project contributors

openlogic:propositional-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The three intermediate facts are asserted inside the counted proof with their justification sources named (induction using prop:phi); the truth-coincidence induction is summarized in the author's own closing sentences. This is the author's complete argument at the level the text intends; the lane accepts citation-driven derivations of this shape (kcomplete/determination precedents).

Editorial note. The truth lemma is NOT delegated to lem:truth (a rejected Exercise-bearing card): the coincidence induction is carried inline in the counted proof.

Editorial note. The label thm:completeness names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Allowed standard prerequisites (editorial).

- Inconsistency is finitely witnessed (one formula at a time). Same unit, \ollabel{prop:phi} referenced as olref[axd]{prop:phi} in the source (source line 485). Line 484-485: justifies fact (1), consistency of each Gamma_n, by induction on n.
- Lindenbaum-style recursive extension. Construction inside the counted proof (lines 463-488): the Gamma_n recursion and Gamma*. The counted proof builds the maximally consistent extension itself; no external card is needed.

Theorem 0.1 (Completeness)**.** *If* Γ *is consistent then* Γ *is satisfiable.*

Proof. Let A_0, A_1, ...be an exhaustive listing of all the formulas of the language. We recursively define an increasing sequence of sets of formulas Γ_0, Γ_1, ..., by putting:

$$\Gamma_0 = \Gamma$$

$$\Gamma_{n+1} = \begin{cases} \Gamma_n \cup \{A_n\} & \text{if } \Gamma_n \cup \{A_n\} \text{ is consistent;} \\ \Gamma_n \cup \{\neg A_n\} & \text{otherwise.} \end{cases}$$

Then define:

$$\Gamma^* = \bigcup_{0 \leq n} \Gamma_n.$$

The proof now proceeds by establishing, in turn, the following facts:

1. For each n, the set Γ_n is consistent (by induction on n, using ??);
2. Γ^* is consistent;
3. Γ^* is maximal.

Then define a valuation v by putting $v(p_i) = \top$ if and only if $p_i \in \Gamma^*$. By induction on A it is then shown that membership in Γ^* coincides with truth according to v also for more complex sentences: $\overline{v}(A) = \top$ if and only if $A \in \Gamma^*$. In particular, v satisfies Γ^*, and since $\Gamma \subseteq \Gamma^*$, also Γ is satisfiable, as desired. $\square$